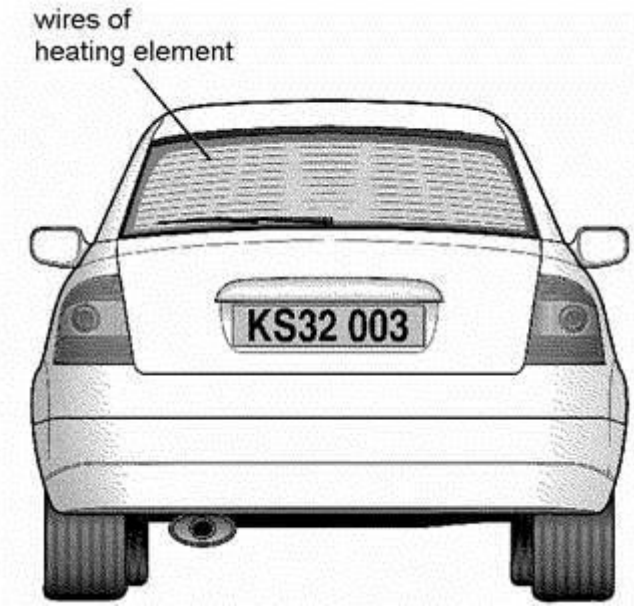
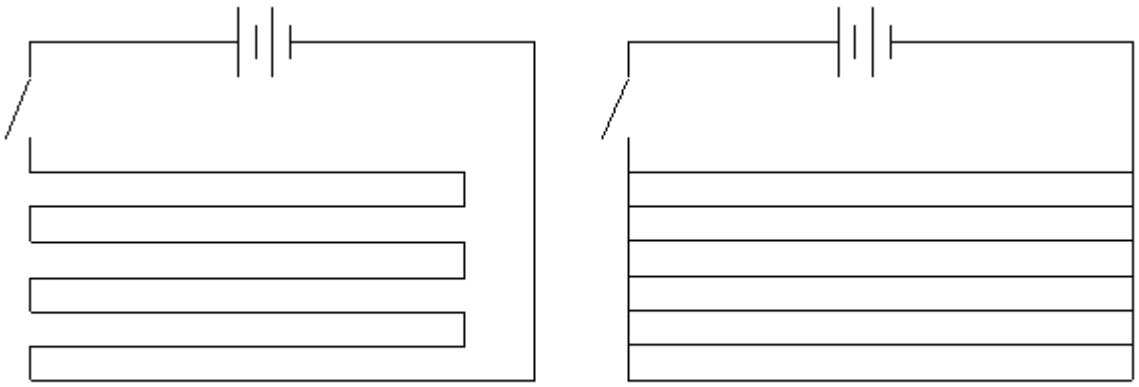
Q1. The back window of this car contains a heating element.

The heating element is part of an electrical circuit connected to the battery of the car.



The diagrams below show **two** ways of connecting the circuit of a heating element.



circuit A **circuit B**

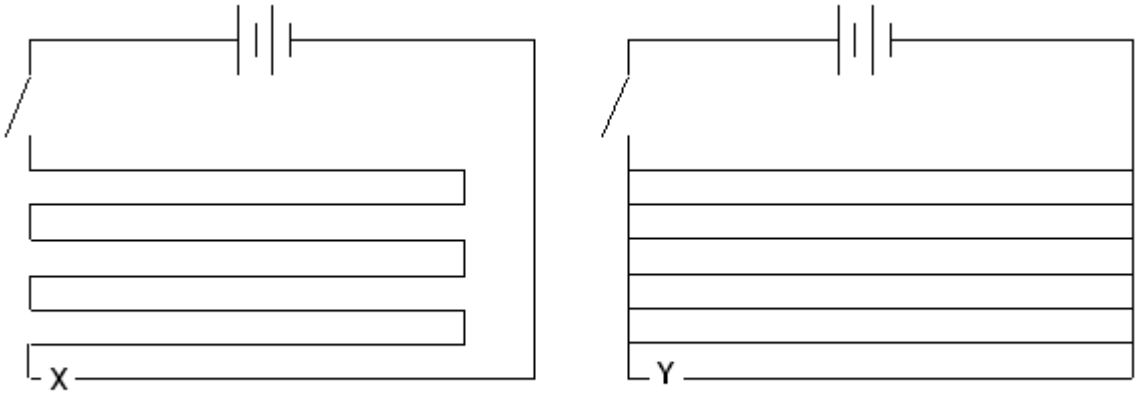
(a) Give the name of each type of circuit:

circuit A .....

circuit B .....

1 mark

(b) A wire gets broken at point X on circuit A and at point Y on circuit B.



circuit A **circuit B**

When the switch is closed, how does the broken wire affect the heating element in:

(i) circuit A? .....

.....

1 mark

(ii) circuit B? .....

.....

1 mark

(c) In very cold weather, ice may form on the back window of the car.
When the heating element is switched on, the ice will disappear and the surface of the window will become clear and dry.

(i) Fill the gap below to show the energy transfer that takes place.

When the heater is switched on, ... energy is transferred from the wires to the ice.

1 mark

(ii) As the window becomes clear and dry, physical changes take place in the ice.
Fill the gaps below to show the physical changes which take place.

from ..... to ..... to .....

1 mark
Maximum 5 marks

Mark schemes

Q1.

(a) *circuit A*: series
circuit B: parallel

***both** answers are required for the mark*

1 (L5)

(b) (i) the circuit **or** heating element will stop working

*accept 'it will not work' **or** 'it will be off'*
accept 'the whole circuit has no current through it'
accept 'it becomes cooler'
*do **not** accept 'it breaks the heater or element or it'*

1 (L5)

(ii) any **one** from

- the circuit **or** element will continue to work
- one wire will not heat the window

accept 'the bottom one has no current through it'
*'nothing' **or** 'it will not be affected' are insufficient*
accept 'it will work less well'
accept 'the bottom wire becomes cooler'
*do **not** accept 'it becomes cooler'*
*do **not** accept 'it does not work properly'*

1 (L5)

(c) (i) thermal accept 'heat'

1 (L5)

(ii) *from* solid *to* liquid *to* gas

***all three** states are required for the mark*
*accept 'from solid to liquid to vapour **or** steam'*
*accept 'from ice to water to vapour **or** gas'*

1 (L5)

[5]